INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

HELIUM LEAK DETECTOR

The General Services Administration has authorized the use of this commercial item description (CID), for all federal agencies.

1. Scope. This CID describes a portable helium leak detector, which will be used to determine the location of leaks in fuel tanks as well as other aircraft systems. By filling a fuel tank or container with helium (the helium acts as a tracer), the helium leak detector will enable aircraft maintainers to locate the origination point(s) of the leak(s) as the helium escapes into the atmosphere.

2. SALIENT CHARACTERISTICS.

2.1 Helium leak detector description. The helium leak detector shall be supplied with a helium leak detector, a standard probe, a heavy duty sampling pump and electronics interface for the safety probe and fast scan probe. The leak detector shall be portable.

2.2 Design and construction. The helium leak detector shall be designed and constructed so that no parts will work loose in service. They shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. They shall be weatherproof and designed to prevent the intrusion of water and sand into critical operating components.

2.2.1 Materials, protective coatings, and finish.

2.2.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable. The use of dissimilar metals only when separated by suitable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur. Sealants or gel type gasket materials shall be used between faying surfaces and butt joints.

2.2.1.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data that may improve this document should be sent to FSC: 6635 - 642 CBSG/GBEA, 460 Richard Ray Blvd, Suite 200, Robins AFB, GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/online/.

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2.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Painted markings shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible, but shall not be less than ½-inch high. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of FED-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of FED-STD-595. Part identification marking is required per MIL-STD-130. Unique identification (UID) is invoked.

2.2.3 Identification and information plates.

2.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the helium leak detector in a readily accessible location. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE) code, date of warranty expiration, and National Stock Number (NSN). The helium leak detector and any components meeting Unique Identification (UID) criteria shall have UID information in accordance with MIL-STD-130 permanently affixed near the respective identification plate(s). UID information shall be included as both a bar code and human readable markings.

2.2.4 Safety.

2.2.4.1 System safety. The design of the helium leak detector shall not contain any system safety mishap risk categories greater than low as defined in Table A-IV of MIL-STD-882.

2.2.4.2 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard.

2.2.4.3 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the helium leak detector with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

2.2.4.4 Electromagnetic interference (EMI). The helium leak detector shall be in accordance with the following radiated emission and susceptibility requirements of MIL-STD-461: RE102 and RS103.

2.2.4.5 Electrostatic discharge. The design of the helium leak detector shall preclude equipment damage due to electrostatic discharge (ESD), protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking.

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2.2.5 Human factors. The helium leak detector shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and Mission-Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear.

2.2.6 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

2.2.7 Welders and welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

2.2.8 Foolproofness. Where improper installation of an item causes a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

2.3 Environmental conditions.

2.3.1 Operating temperature range. The helium leak detector shall be capable of operating in ambient temperatures ranging from -68 F to 104 F.

2.4 Transportability.

2.4.1 Surface transportability. The helium leak detector shall be transportable via all modes of surface shipment: rail, sea, road, etc., and shall be capable of withstanding mechanical shock and vibration characteristics of rail, sea, and road transport.

2.5 Performance.

2.5.1 Concentration. The minimum helium detectable concentration required shall be two parts per million.

2.5.2 Detectable leak rate. The minimum detectable leak rate required shall be $5x10^{6}$ millibar liters per second.

2.5.3 Response time. The leak detector shall have a response time of less than two seconds.

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2.6 Probes.

2.6.1 Standard probe. The detector shall be supplied with a standard probe that is supplied with the detector.

2.6.2 Standard probe hose length. The standard probe hose length shall be at least three feet.

2.6.3 Safety probe. A safety testing probe shall be supplied that has been certified intrinsically safe to National Fire Protection Association 70 Class 1 Division 1 hazardous areas requirements.

2.6.4 Certificate of conformance. The safety testing probe shall be supplied with a copy of the Class 1 Division 1 certificate of conformance.

2.6.5 Safety probe hose length. The safety probe hose length shall be at least 30 feet.

2.6.6 Non-safety probe. The detector shall be supplied with a non-safety telescoping detector probe for leak detection in non-fuel source related applications.

2.6.7 Non-safety probe hose length. The length of the non-safety probe shall be capable of telescoping at least 10 feet.

2.6.8 Pin probe. The pin probe shall be supplied with a pin probe kit, which includes three attachments to allow detection in restricted access areas.

2.6.9 Straight metallic probe. The length of the straight metallic probe shall be six inches.

2.6.10 Flexible plastic probe. The length of the flexible plastic probe shall be six inches.

2.6.11 Oil can gooseneck probe. The length of the oil can gooseneck probe shall be six inches.

2.6.12 Electronic probe interface. The helium leak detector shall be supplied with an electronic probe interface, which allows the connection of the safety probe, and the fast scan probe to the detector unit.

2.7 Heavy duty sampling pump. The heavy duty sampling pump is an air pump that draws the air being sampled into the detector in order for the detector to “sniff” the sampled air for helium.

2.8 Helium manifold network. The helium manifold network is designed to allow regulated low pressure helium [less than five per square inch (PSI)] to be applied to the aircraft.

2.8.1 Digital pressure gauges. Two digital pressure gauges that are battery powered and have an operating range of 0 to 240 PSI.

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2.8.2 Dual position flow control valve. One dual position flow control valve shall be required to control the current range of output pressure (0 to three PSI or 0 to 70 PSI range). The dual position flow control valve determines which range is currently being used.

2.8.2.1 Position 1. Zero to three PSI (left hand facing position).

2.8.2.2 Position 2. Zero to 70 PSI (right hand facing position).

2.8.3 Valves. Two regulator valves shall be included in the helium leak detector kit. The helium leak detector kit consists of the following: ground wire, helium bottle belt, helium connector, air connector, air filling hose, measuring hose, purge hose, injection needle, power cord, bagging material, helium regulator, helium hose, injection hose, sticky tape, and aluminum tape. The left hand valve controls the 0 to three PSI range of helium pressure. The right hand valve controls the 0 to 70 PSI range of helium pressure. The 0 to three PSI pressure is measured by the left hand gauge and the 0 to 70 PSI range is measured by the right hand gauge.

2.9 Bonding strap. The helium leak detector shall be supplied with a bonding strap that electronically bonds the helium leak detector kit with the aircraft that is being tested.

2.10 Connector tubes. The helium leak detector shall be supplied with all necessary connector tubes to allow the helium leak detector kit to be connected to the helium supply as well as the manifold network to the aircraft.

2.11 Input voltage source. The leak detector shall operate at 110 or 220 VAC, 50 to 60 Hz, as well as 12 volt direct current (VDC) at 1.2 A using a rechargeable Ni-MH battery or equivalent.

2.12 Batteries. Two batteries shall be supplied and must be a commercially available nickel metal hydride (Ni-MH) battery (or equivalent) that is commercially available.

2.12.1 Battery charger. The helium leak detector shall be supplied with two battery chargers to charge its batteries.

2.13 Foreign power requirements. If the solicitation indicates destinations outside of the United States then the offered items shall have the proper power requirements and electric plugs or adapters for the destinations indicated.

2.14 Operating conditions.

2.14.1 Relative humidity. The helium leak detector shall be capable of operating at a maximum relative humidity of 90%.

2.15 Detector dimensions. The dimensions of the helium leak detector shall be approximately 8” x 7” x 4”.

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2.16 Detector weight. The helium leak detector shall not weigh more than 10 pounds including the battery.

2.17 System weight. The helium leak detector kit shall not weigh more than 100 pounds including all components.

2.18 Leak detector capability/hardware. All equipment required for each test shall be supplied with the helium leak detector, unless otherwise indicated. A helium source is not required.

2.19 Storage/carrying case. The helium leak detector shall be supplied with a heavy duty storage/carrying case capable of storing the detector and all related components including the operation manual. The case shall be supplied with wheels and two carrying straps for easy transporting of the system.

2.20 Workmanship. The helium leak detector, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

2.20.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

2.20.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list

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that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4.0 PRODUCT CONFORMANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of the CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be appropriate. Proof of conformance may include, but shall not be limited to the performance of operational tests, lab tests, modeling and simulation and delivery of reports and data from these tests.

4.2 Installation and demonstration. The contractor shall be responsible for user orientation of the equipment at the designated Air Force facility and to demonstrate that the equipment is ready for use. Installation and demonstration is projected to take three days. The time shall be divided between classroom demonstration/orientation and on the job training with the users.

5. PACKAGING.

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Source of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at https://assist.daps.dla.mil/online/ or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/ . Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dpap/dars/dfars/index.htm .

6.1.3 National Fire Protection Association (NFPA)/National Electrical Code (NEC) standards may be obtained at http://www.nfpa.org or available from 1 Batterymarch Park, Quincy, Massachusetts USA 02169-7471

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6.2 Intended use. The purpose of the helium leak detector is to locate small leaks that are often found within tanks or containers. This will allow the aircraft maintainers to take corrective action and patch or fill the leaking areas.

6.3 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this commercial item description.

6.4 Key Words

Aircraft
Fuel tanks
Portable
Tracer

MILITARY INTERESTS:

Custodians:
Air Force – 84

Preparing Activity:
Air Force – 84

Reviewers:
Air Force – 99

Agent:
Air Force – 99

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